ASSIGNMENT 3 Coastal Engineering

Assessment on Seawalls as a Coastal Engineering Intervention

ENCN442-26S1

Surface Water, Ground Water and Coastal Engineering

Zac Hemmings

zhe32@uclive.ac.nz

University of Canterbury

Date Submitted: 13 May 2026

Generative AI was used to assist with editing and formatting in this report in accordance with the ENCN442 course guidelines. In this case, use of an LLM was requested to begin with. All prompts have been included.

Assignment Duration about 8 Hours

SUMMARY

The initial prompt was simple and noted some key areas for the LLM to work on. The LLM did a great job at outlining the various areas and forming a structure for the report, but some areas were slightly misaligned, and there were grey areas that had to be backed up with a reference.

It seems the LLM got the theory for implications of seawall's mostly correct, although it made some bold statements in regards to the cost and implications of seawalls, speaking generically when seawalls have many different forms. The LLM mentioned Maori cultural values, but didn't fully account for the treaty of Waitangi principles, and actually clearly defining what the principles mean in terms of seawalls. The LLM however failed to consider bicultural considerations such as the perspectives of other ethnic groups, besides Maori. It did a reasonable job explaining the values of kaitiakitanga and protecting natural resources, but to fully understand perspectives from a bicultural approach, we would need to consider a range of other perspectives.

Initial Prompt: *"Generate 750 words on seawalls as a coastal engineering intervention, im interested on a bit of design characteristics, effects on quasi steady-state, key engineering aspects, cultural considerations and climate change impacts"*

1. LLM OUTPUT

Seawalls are one of the most common coastal engineering interventions used to protect land, infrastructure, and communities from waves, storm surge, and coastal erosion. They are built along the shoreline and act as a hard barrier between the sea and the land. In most cases, seawalls are used in urban or developed coastal areas where assets such as roads, housing, ports, and public services are too valuable or difficult to relocate. They can provide effective short-term protection against erosion and flooding, but they also alter natural coastal processes and can create environmental, geomorphological, and cultural impacts that must be considered during design and planning.

There are several main types of seawalls used depending on site conditions and wave climate. Vertical seawalls are rigid concrete structures that reflect wave energy back toward the ocean and are typically used where space is limited. Curved or recurved seawalls include a lip that redirects wave energy upward and helps reduce overtopping during storms. Rubble mound seawalls are sloped structures made from large rock armour units that dissipate wave energy through friction and turbulence rather than direct reflection. In some locations, hybrid approaches are also being explored, combining traditional seawalls with vegetation or ecological features to improve environmental outcomes. The choice of seawall depends on factors such as wave exposure, sediment conditions, available construction materials, and the level of protection required.

From a coastal processes perspective, seawalls significantly affect the natural quasi steady-state behaviour of beaches. Under natural conditions, beaches continuously adjust to waves, tides, and sediment movement, tending toward a dynamic equilibrium over time. When a seawall is built, it fixes the shoreline position and prevents natural landward migration of the coast. This often increases wave reflection, causing greater turbulence and scour at the base of the structure. Over time, this can narrow the beach in front of the seawall and reduce sediment volume. Beaches may become steeper and less stable, and in some cases disappear entirely in front of the structure. This process is often referred to as coastal squeeze, where the beach becomes trapped between a fixed hard structure and rising sea levels. Seawalls can also interrupt longshore sediment transport, potentially shifting erosion problems further along the coast.

The design of seawalls requires careful consideration of several engineering factors. Wave loading is critical, including both everyday wave conditions and extreme storm events. Engineers must also account for storm surge levels, wave run-up, overtopping, and future sea level rise projections. Foundation stability is especially important because scour at the toe of the structure can undermine the wall and lead to failure. Toe protection such as rock aprons or sheet piling is therefore commonly included. Materials must also withstand long-term saltwater exposure, corrosion, and repeated wave impact. Overtopping remains a key issue, particularly as climate change increases the likelihood of coastal flooding. Even if a seawall remains structurally stable, water may still pass over the wall during extreme events and damage assets behind it. Seawalls are generally expensive to construct and maintain, requiring regular inspection and repair throughout their design life.

Cultural considerations are particularly important in coastal engineering in New Zealand. Coastal environments are not only physical systems but also hold deep cultural, spiritual, and historical significance for Māori. Concepts such as kaitiakitanga emphasise guardianship and long-term stewardship of the natural environment. Coastal areas may also contain wāhi tapu, urupā, and mahinga kai, meaning they are closely connected to identity, heritage, and customary food gathering. Seawalls can sometimes conflict with these values by hardening natural coastlines, reducing access to beaches, and altering natural character. They may also affect the mauri of coastal systems by disrupting natural coastal processes. Because of this, iwi and hapū consultation is an important part of coastal planning and design. The Resource Management Act 1991 requires consideration of Māori relationships with ancestral lands and water, while the New Zealand Coastal Policy Statement 2010 encourages involvement of tangata whenua in coastal decision-making and protection of natural coastal character. As a result, softer approaches such as dune restoration, beach nourishment, or managed retreat are increasingly considered alongside hard engineering structures like seawalls.

Climate change is placing increasing pressure on coastal protection systems, including seawalls. Rising sea levels, changing wave climates, and more intense coastal storms mean many existing seawalls may become less effective over time. Structures designed using historical climate conditions may require future upgrades or reinforcement as coastal hazards intensify. There is also concern that seawalls can create a false sense of security, encouraging continued development in vulnerable coastal areas. Because of this, coastal engineering is increasingly shifting toward adaptive management approaches that combine traditional hard engineering with nature-based solutions and long-term planning strategies. Overall, seawalls remain an important coastal protection tool, but they are increasingly viewed as only one part of a broader and more adaptive approach to coastal management under climate change.

2. SEAWALLS AS A COASTAL ENGINEERING INTERVENTION

Seawalls are one of the most recognisable forms of coastal engineering intervention, designed to protect coastal land, infrastructure, and communities from erosion and wave attack. They are typically constructed parallel to the shoreline and act as a hard barrier between the sea and the land. Seawalls are commonly used in urban coastal environments where there is high-value infrastructure such as roads, ports, housing, or public facilities that require protection from storm surge and long-term shoreline retreat. While seawalls can be highly effective in the short term (Sanitwong-Na-Ayutthaya et al. 2023), they also create significant environmental, geomorphological, and cultural consequences that must be considered in modern coastal engineering practice (Blackett 2007).

There are several main types of seawalls, each designed for different coastal conditions and wave climates. Vertical seawalls are steep structures, often made from reinforced concrete, that reflect incoming wave energy directly back toward the ocean. These are commonly used in heavily urbanised coastlines because they occupy relatively little space. Curved or recurved seawalls include an overhanging lip that redirects wave energy upward and back seaward, reducing overtopping during storms. Rubble mound seawalls, sometimes referred to as revetments, consist of large rock armour units placed on a slope. These structures dissipate wave energy through friction and turbulence rather than reflecting it entirely, generally making them more stable and environmentally compatible (El-abosy et al. 2020). In recent years, engineers have also experimented with hybrid and “living shoreline” approaches that combine seawalls with vegetation or ecological features to improve resilience and biodiversity outcomes (Minuti 2026).

The design of seawalls depends heavily on site-specific coastal processes. Engineers must consider wave height, wave period, tidal range, sediment transport, storm frequency, and projected sea level rise. Foundation stability is critical because undermining from scour can lead to structural failure. Toe protection is therefore commonly included to reduce erosion at the base of the wall. Materials must also withstand long-term exposure to saltwater corrosion, hydraulic pressure, and impact loading from waves and debris. Overtopping is another key consideration, particularly as climate change increases the frequency of extreme coastal storms. Even if a seawall remains structurally intact, overtopping can still cause flooding and damage to assets behind the wall (McGlade et al. 2025).

Highlighted: This is also contested in the article by McGlade where he cites another article by Jennath and Paul, 2024 “Climate change is increasing the frequency and magnitude of coastal flooding and wave overtopping”. This should be investigated further as storm intensity is increasing, but frequency of events vary by region and are not as certain.

One important coastal engineering concept associated with seawalls is the quasi-steady state of a beach system. Under natural conditions, coastlines tend to adjust dynamically toward an equilibrium profile where sediment transport processes are balanced over time. Beaches naturally shift landward or seaward in response to storms, tides, and changing sediment supply. The construction of a seawall interrupts this natural adjustment process by fixing the position of the shoreline. While the seawall may initially prevent erosion of the land behind it, the beach in front of the structure often narrows over time (Pilkey and Cooper 2012). This occurs because wave reflection from the wall increases turbulence and scour, accelerating sediment loss near the structure (French 2001). Eventually, the beach can enter a new quasi-steady state characterised by a steeper beach profile and reduced sediment volume.

This process is sometimes referred to as "coastal squeeze," where the beach becomes trapped between rising sea levels and the immovable seawall. As sea level rises, the beach cannot naturally migrate inland, resulting in progressive beach loss (Lester 2017). This creates both ecological and recreational impacts, particularly in areas where beaches are valued for tourism or habitat. In some cases, seawalls can even transfer erosion problems further along the coast by disrupting longshore sediment transport processes (Angnuureng et al. 2025). This demonstrates how coastal interventions often solve one issue locally while generating unintended consequences elsewhere in the coastal system.

From an engineering perspective, seawalls are often justified where the value of protected assets outweighs environmental or construction costs. In densely populated coastal cities, retreating inland may not be economically or socially feasible. However, seawalls are extremely expensive to construct and maintain over their design life. As climate change intensifies coastal hazards, existing seawalls may require continual upgrading to maintain acceptable levels of protection. Engineers are increasingly recognising that seawalls alone may not represent a sustainable long-term solution, particularly in areas exposed to accelerating sea level rise (Abd-Elaty 2025).

Highlighted: This statement seems unreasonable, as seawalls are diverse in nature, each requiring varying levels of design. Some rock constructed seawalls are relatively inexpensive, compared to tall engineered seawalls for example.

Cultural considerations are also highly important in coastal engineering projects. In New Zealand, Māori perspectives such as kaitiakitanga emphasise the guardianship and long-term stewardship of the coastal environment. Hard engineering structures like seawalls can conflict with these values if they damage natural ecosystems, restrict public access, or alter culturally significant landscapes (Bailey-Winiata 2026). Coastal areas may also contain wāhi tapu sites or hold strong cultural and spiritual importance for tangata whenua. As a result, modern coastal engineering increasingly involves consultation and co-design with local iwi and hapū to ensure that interventions align with both technical objectives and cultural values. In some cases, softer adaptation approaches such as dune restoration or managed retreat may better support both ecological and cultural outcomes (Lawrence et al. 2025). c

In New Zealand, cultural and bicultural considerations are central to seawall and coastal engineering decisions, specifrically kaitiakitanga, iwi engagement, and Te Tiriti o Waitangi obligations. Coastal areas are not only physical systems but also hold deep cultural and spiritual meaning for Māori, including as a food cource, and preserving treasured land (Taonga). Because of this, seawalls can do more than change wave behaviour and sediment transport, they can also restrict access to the coast, a resource highly prized bv Maori culture, damage natural character, and disrupt relationships between people and place if they are not carefully designed with local communities accounted for. Under the Resource Management Act 1991, councils must recognise Māori relationships with ancestral lands and water, give regard to kaitiakitanga, and take the treaty principles into account, which makes iwi consultation a required part of coastal projects. The New Zealand Coastal Policy Statement 2010 also pushes for early engagement with tangata whenua and prioritises protecting natural coastal character, often avoiding hard structures that modify the natural state of the land. It is therefore important to involve iwi early in design and planning, and there is a growing shift toward solutions that better align with kaitiakitanga such as dune restoration, managed retreat, or softer hybrid protections instead of just seawalls (Department of Conservation 2010; Ministry for the Environment 1991).

Climate change presents a major challenge for the future viability of seawalls. Rising sea levels, increased storm intensity, and changing wave climates will place greater pressure on coastal protection structures. Many seawalls designed using historical climate data may become inadequate within coming decades. There is also growing concern that reliance on seawalls may create a false sense of security, encouraging further development in vulnerable coastal areas (Stelljes 2025). Consequently, coastal engineers are increasingly shifting toward adaptive management approaches that combine engineering protection with long-term planning, hazard zoning, and ecosystem-based adaptation strategies.

3. REFLECTION ON LLM USE IN ENGINEERING

Overall the LLM provided a comprehensive, structured report on seawalls in a short time frame. For assessing the general characteristics of a feature like this, the report produced by the LLM would be suitable. Most of the wording was correct, with a few small conceptual errors and lack of depth, which comes from the LLM not actually researching dep enough into articles about some of these features. I would recommend that LLM's are useful to engineers to broaden their thinking, potentially outlining different areas to consider. Where the LLM falls short is to actually ask people for their perspective. When dealing with local iwi and their land, it is important to actually ask them how they feel about your design, and whether they think there will be any issues. It is not just to ask if it aligns with their values, but the act of consulting them implies that the engineer cares for their perspective and develops a positive relationship with them. The output should be read through thoroughly by the engineer, ensuring to hold their own perspectives and knowledge over the wording of the LLM.

REFERENCES

Abd-Elaty, I. (2025). "Subsurface seawalls for sustainable coastal aquifers and inundation prevention under climate change and tidal waves impact." *Journal of Coastal Research*, Taylor & Francis.

Angnuureng, D. B., Charuka, B., Almar, R., Dada, O. A., Asumadu, R., Agboli, N. A., and Ofosu, G. T. (2025). "Challenges and lessons learned from global coastal erosion protection strategies." *iScience*, 112055. https://doi.org/10.1016/j.isci.2025.112055

Bailey-Winiata, A. P. S. (2026). *Understanding the potential exposure of coastal marae and urupā in Aotearoa New Zealand to sea level rise*. University of Waikato, Hamilton, New Zealand.

Blackett, P. (2007). *Community involvement in coastal hazard mitigation: Some insights into process and pitfalls*. West Coast Regional Council, New Zealand. https://www.wcrc.govt.nz/repository/libraries/id:2459ikxj617q9ser65rr/hierarchy/Documents/Publications/Natural%20Hazard%20Reports/West%20Coast/Community%20Involvement%20Paper%20T%20Hume%20NIWA%202007.pdf

El-abosy, M., Farag, O., and Moustafa, W. (2020). "Wave Hydrodynamic Characteristics of Vertical and Sloped Seawalls." *Bulletin of the Faculty of Engineering, Mansoura University*, 40(12-20). https://doi.org/10.21608/bfemu.2020.101057

French, P. W. (2001). *Coastal Defences: Processes, Problems and Solutions*. Routledge, London, UK.

Lawrence, J., Gallop, S. L., Marquardt, L., Bell, R., Blackett, P., Cradock-Henry, N. A., and Wreford, A. (2025). "Dynamic adaptive pathways planning for adaptation: Lessons learned from a decade of practice in Aotearoa New Zealand." *Journal of Integrative Environmental Sciences*, 22. https://researcharchive.lincoln.ac.nz/server/api/core/bitstreams/8f993414-ba0c-4ad0-997a-9e0361b4f521/content

Lester, C. (2017). "Managing the coastal squeeze: Resilience planning for shoreline residential development." *Stanford Environmental Law Journal*, 36(2), 24. https://law.stanford.edu/wp-content/uploads/2017/11/lester.pdf

McGlade, M., Valiente, N. G., Brown, J., Stokes, C., and Poate, T. (2025). "Investigating appropriate artificial intelligence approaches to reliably predict coastal wave overtopping and identify process contributions." *Ocean Modelling*, 194, 102510. https://doi.org/10.1016/j.ocemod.2025.102510

Minuti, J. J. (2026). "Balancing urbanization and ocean sustainability: the evolution of eco-shorelines through cross-disciplinary collaboration in Hong Kong." *Frontiers in Ocean Sustainability*.

Nielsen, A. (2026). "Wave energy considerations for the concept design of rock armored coastal revetments." *LJER*, Great Britain Journals Press. https://www.journalspress.uk/index.php/LJER/article/download/1754/4561

Pilkey, O. H., and Cooper, J. A. G. (2012). *The Last Beach*. Duke University Press, Durham, NC.

Sanitwong-Na-Ayutthaya, S., Saengsupavanich, C., Ariffin, E. H., Ratnayake, A. S., and Yun, L. S. (2023). "Environmental impacts of shore revetment." *Heliyon*, 9(9), e19646. https://doi.org/10.1016/j.heliyun.2023.e19646

Stelljes, N. (2025). *Ecosystem-based adaptation for coastal regions worldwide*. Ecologic Institute, Berlin, Germany. https://www.ecologic.eu/sites/default/files/event/2025/TropcialAdapt-Background-paper.pdf

Sun, H., Xu, J., Zhang, S., Li, G., Liu, S., Qiao, L., Yu, Y., and Liu, X. (2022). "Field observations of seabed scour dynamics in front of a seawall during winter gales." *Frontiers in Marine Science*, 9, 1080578. https://doi.org/10.3389/fmars.2022.1080578

Department of Conservation (DOC). 2010. New Zealand Coastal Policy Statement 2010. Wellington, New Zealand.

Ministry for the Environment (MfE). 1991. Resource Management Act 1991. Wellington, New Zealand.